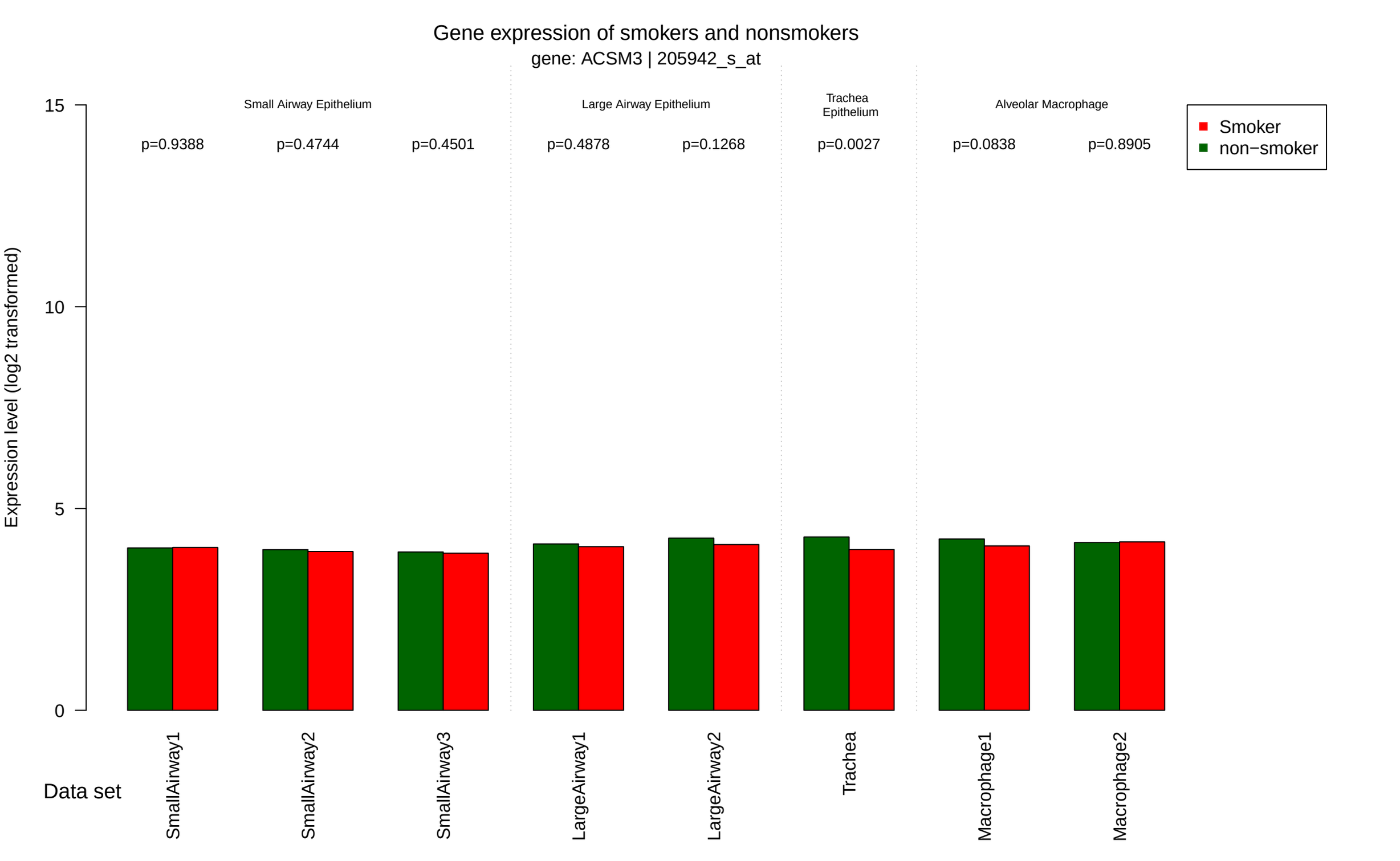
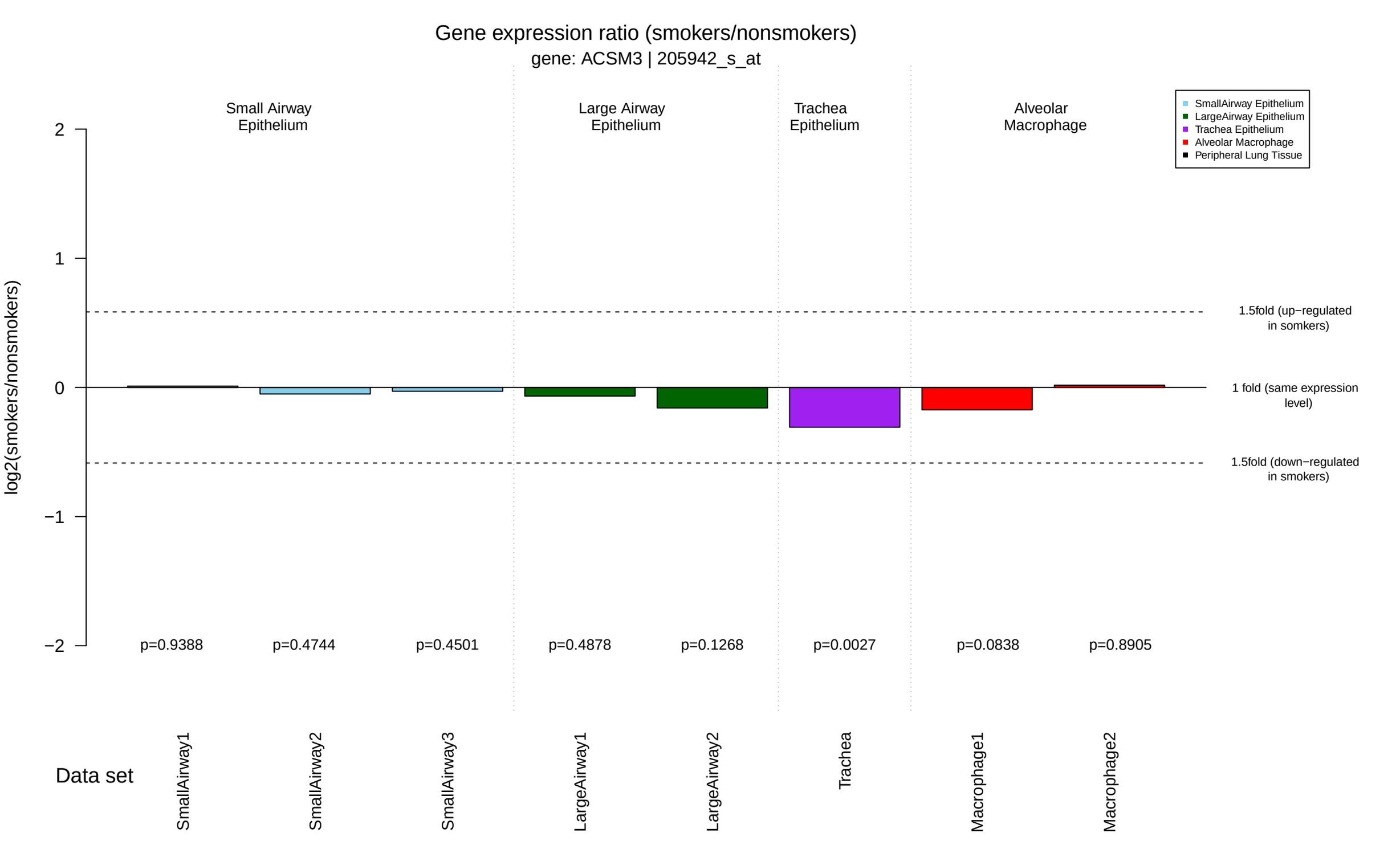
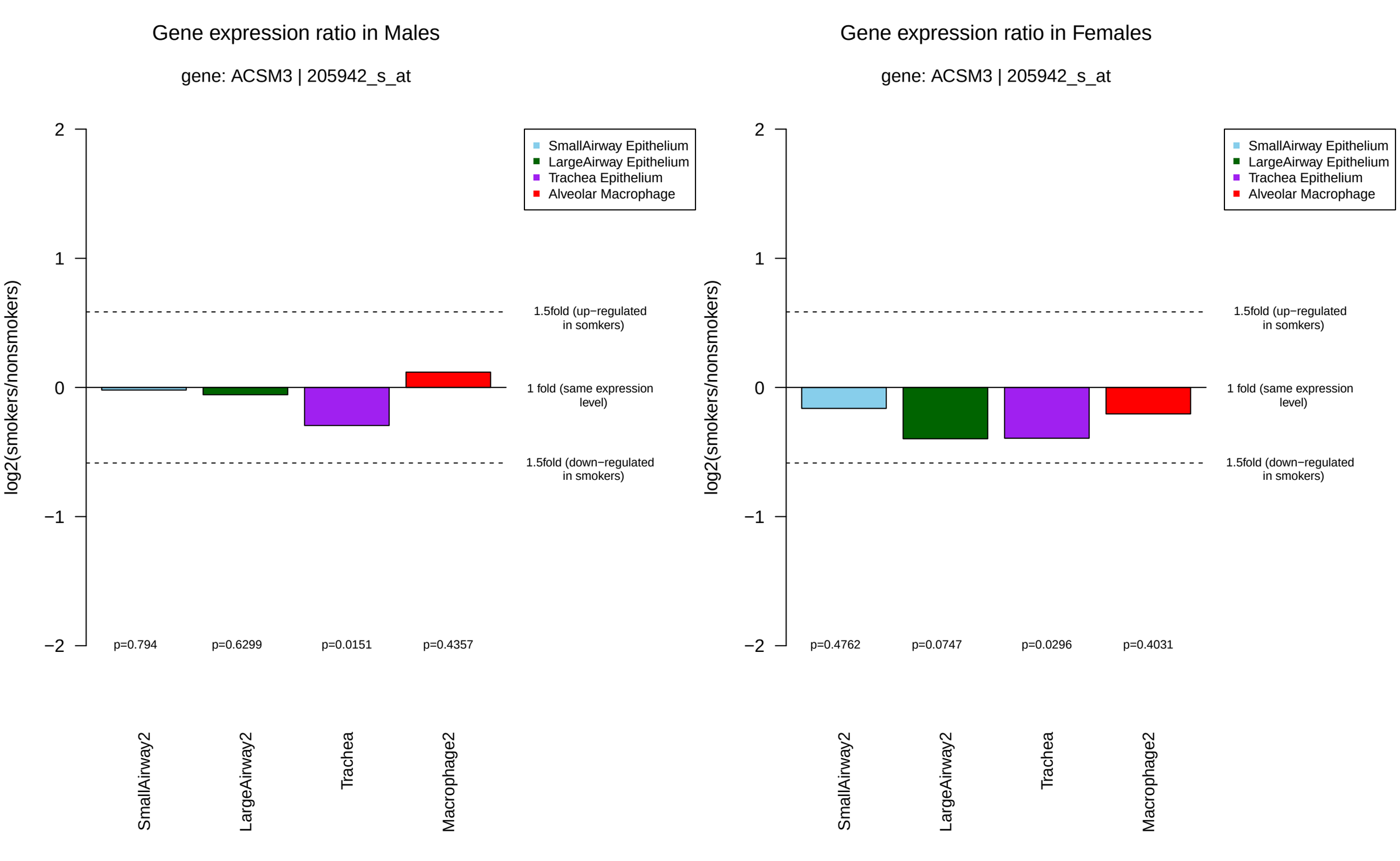
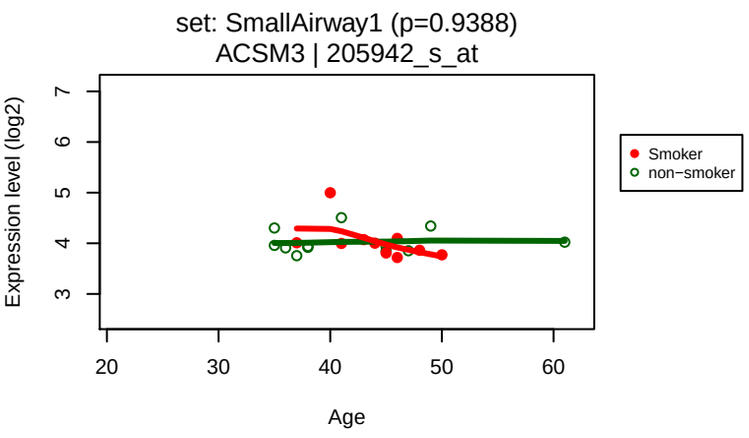
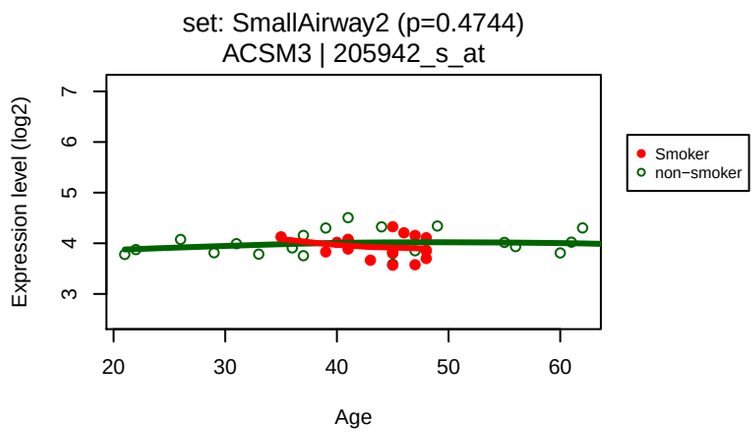
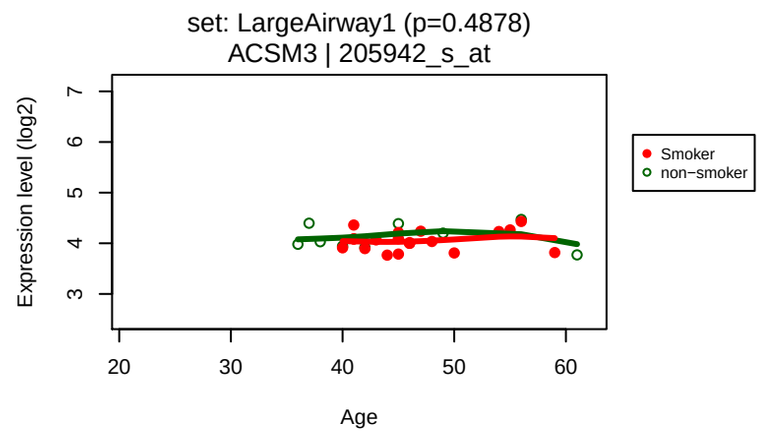

Gene expression of smokers and nonsmokers
gene: ACSM3 | 205942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9388
p=0.4744
p=0.4501
p=0.4878
p=0.1268
p=0.0027
p=0.0838
p=0.8905
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

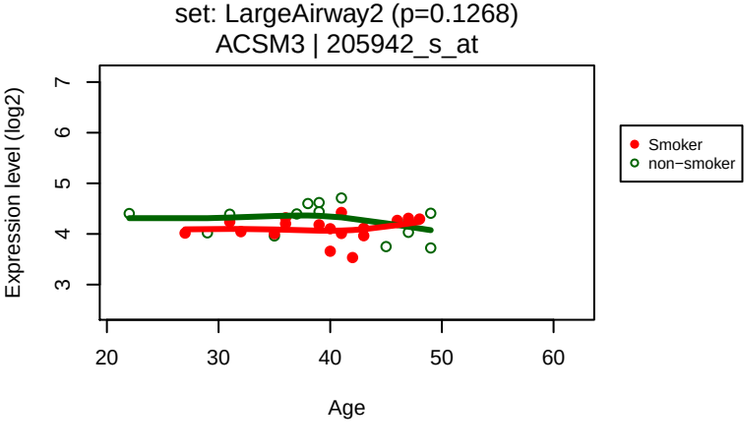

Gene expression ratio (smokers/nonsmokers)
gene: ACSM3 | 205942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9388
p=0.4744
p=0.4501
p=0.4878
p=0.1268
p=0.0027
p=0.0838
p=0.8905
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

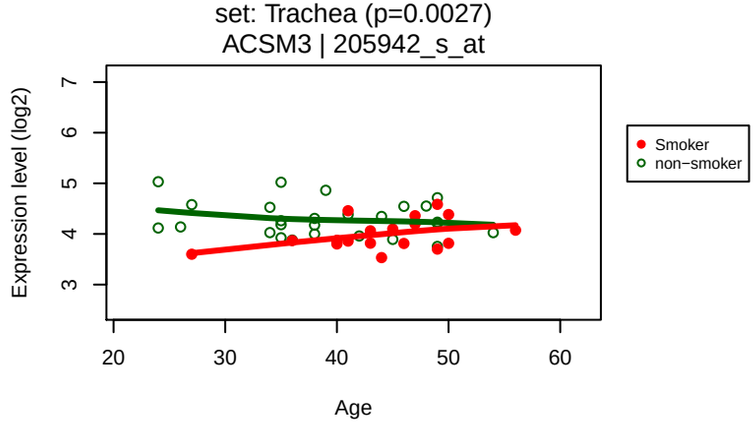

Gene expression ratio in Males
gene: ACSM3 | 205942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.794
p=0.6299
p=0.0151
p=0.4357
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSM3 | 205942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4762
p=0.0747
p=0.0296
p=0.4031
SmallAirway2
LargeAirway2
Trachea
Macrophage2

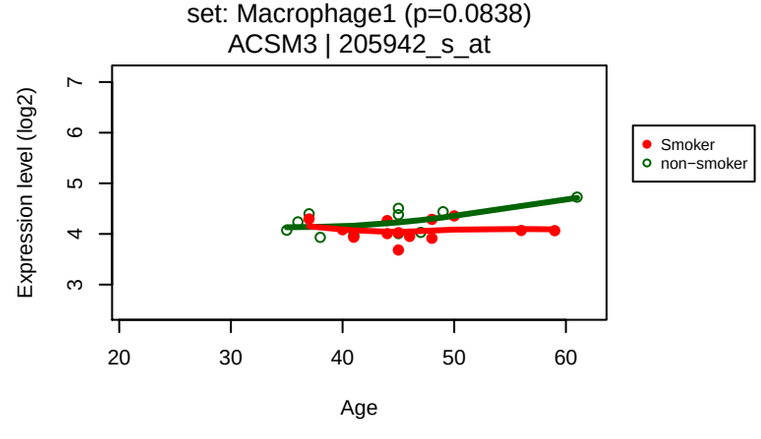

set: SmallAirway1 (p=0.9388)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker

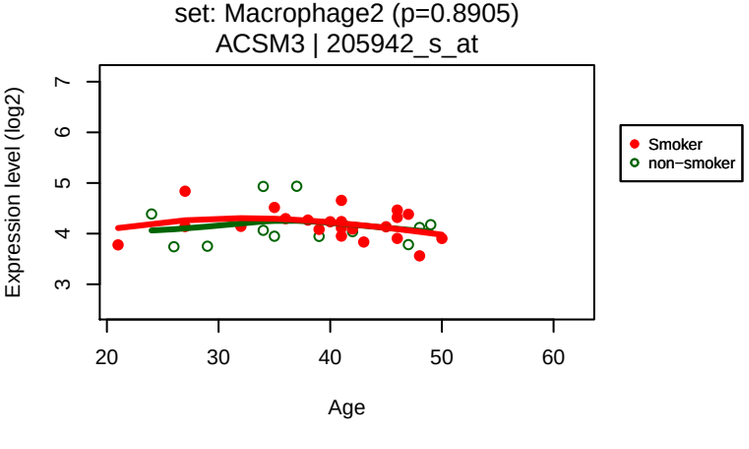

set: SmallAirway2 (p=0.4744)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker

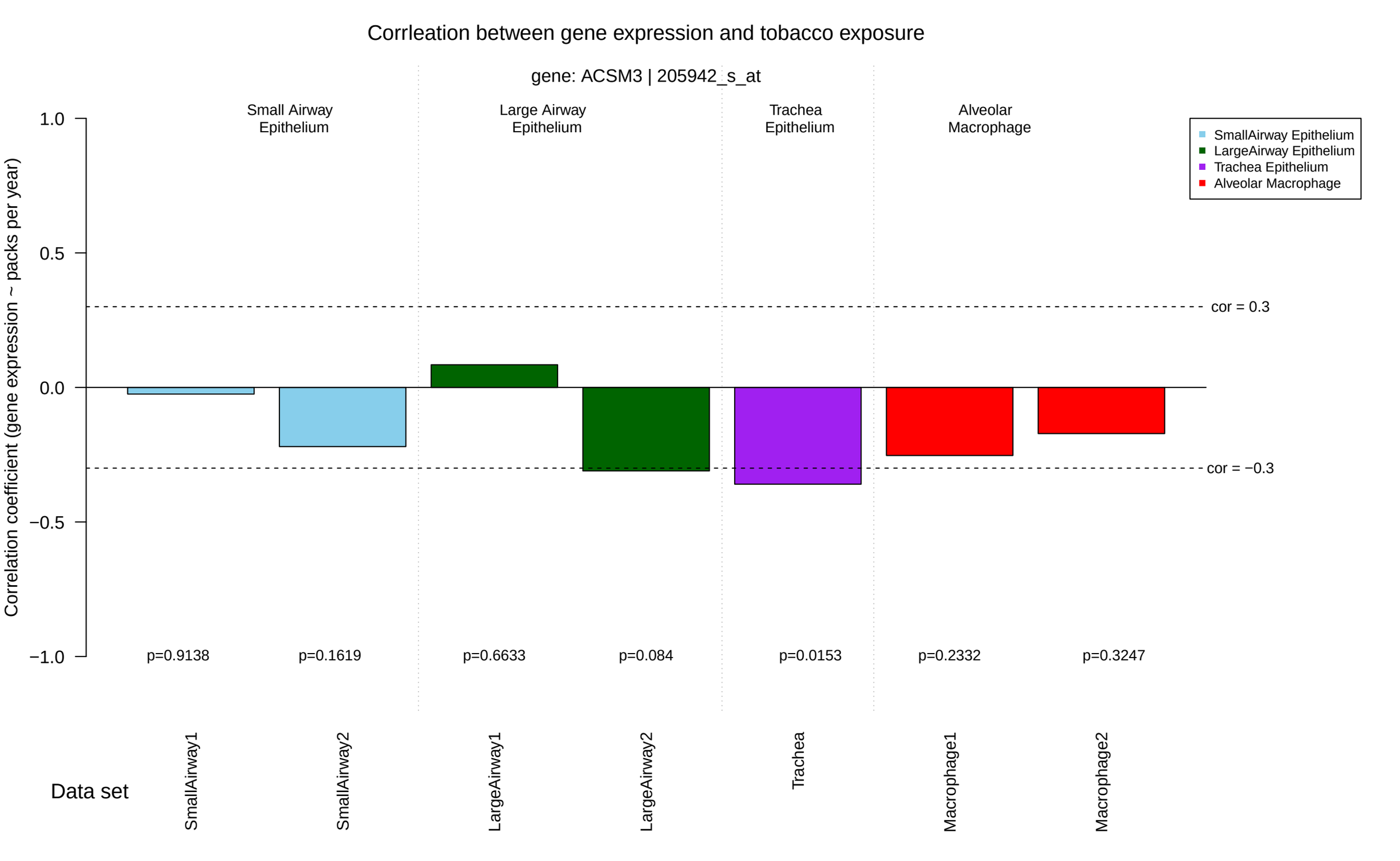

set: LargeAirway1 (p=0.4878)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1268)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0027)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0838)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8905)
ACSM3 | 205942_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSM3 | 205942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9138
p=0.1619
p=0.6633
p=0.084
p=0.0153
p=0.2332
p=0.3247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2